

Ethan Kennedy

Full-Stack & AI Engineer

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Profile

Software and AI engineer with **5+ years** of hands-on experience building full-stack systems and shipping AI-powered products to production. Joined Continuata before starting a CS degree at Glasgow, so commercial engineering has run in parallel with formal study throughout. Draws to AI engineering for the systems challenges: context management, tool design, evaluation, and cost-aware routing. Strong in Cloudflare edge infrastructure and LLM systems. Graduating June 2026.

Experience

Continuata Services Ltd continuata.io *2021 – Present*
Full-Stack Engineer

- Built a complete FastSpring integration: webhook handler, merchant dashboard UI, and documentation from scratch.
- Fixed critical bugs in the v3 download pipeline — hashing performance, upload timeouts on large folders, and silent download truncation — contributing to observed 500 Mbps+ client throughput in production.
- Shipped landing page redesign, pricing calculator with competitor comparison, and a full documentation overhaul for the V3 pack-based architecture.
- Contributed fixes to the Desktop Bridge (Go) — HTTPS auto-cert generation, download speed reporting, and integration test coverage.

University of Glasgow *2025 – 2026*
Software Development Team Coach

- Coached a third-year software engineering team through a full development lifecycle project — from requirements through delivery.
- Mentored students on Agile practices, collaborative workflows, and rapid adoption of unfamiliar technologies.
- Facilitated workshops following industry-led training delivered by JP Morgan engineers.

Freelance Software & Creative Work *2020 – Present*

- Produced and mixed music for 80+ artists, contributing to projects with 500M+ Spotify streams.
- Recent: **TM Trees** (tmtrees.com) — production marketing site on Cloudflare Workers (Hono, TypeScript, Tailwind, daisyUI). Shipped contact-form pipeline (Resend API, Cloudflare Turnstile CAPTCHA), Google Drive integration that auto-syncs the media gallery on file drop with no redeploy, SEO restructure, and live Google Reviews embed.
- Voiceover work for animation, video games, audiobooks, and YouTube channels.

Selected Projects

A.R.I.A. — Audio Routing & Instrumentation Assistant

- Real-time AI voice agent embedded inside a DAW: voice-driven session setup, take management, FX-chain templating, and mix assistance — an AI system orchestrating ~60 tools via natural language.
- Tauri 2 desktop client + Pipecat voice pipeline (Deepgram → GPT → Cartesia), bridged over file-IPC to a ReaScript Lua tool surface inside Reaper.
- Fuzzy plugin matching, cross-format scanning (VST/VST3/AU/JSFX), and role-based FX templates with automatic stock-plugin fallbacks.

LLM Inference Platform

- Production LLM gateway on Cloudflare Workers: authenticated chat (JWT/JWKS), router-based model selection, AI Gateway integration, normalised SSE streaming, and automatic fallback between Workers AI and external providers.
- Durable Object rate limiting with budget enforcement, a versioned prompt registry, and Analytics Engine cost/latency observability surfaced via Prometheus and Grafana.
- Terraform IaC, multi-environment continuous deployment with smoke tests, and scheduled production verification with alerting.

Real-Time ML Platform — Streaming Fraud Detection

- End-to-end MLOps stack: PyFlink streaming features → Feast/Redis online store → Ray Serve + ONNX Runtime → MLflow registry with shadow deployment, drift detection (KS + PSI), and one-click rollback.
- 231 ms p95 / 174 RPS on a local benchmark; pre-warmed ONNX session pool, msgspec-typed decoding, and bounded concurrency.
- Online adaptation via SGD micro-batches and a LinUCB contextual bandit; Evidently drift triggers gated retraining.

Yorùbá Learning PWA

- Spaced-repetition language learning for a low-resource language: FSRS algorithm, frequency-ordered phrase deck (Tatoeba + Masakhane), CEFR-style fluency ranks.
- On-device pronunciation grading via HuggingFace MMS-ASR; native audio generated with MMS-TTS and stored in Cloudflare R2.
- Stack: Hono, Cloudflare Workers, D1, Drizzle, ts-fsrs, HTMX, Tailwind.

MIDI-to-Piano Neural Synthesis — Honours Dissertation

- Hybrid physics + neural piano synthesizer: ModalPianoV7 (40-mode differentiable modal synthesis with inharmonicity, body convolution, and sympathetic resonance) and GAN V3 (U-Net generator + PatchGAN discriminator for spectrogram style transfer).
- Multi-resolution STFT, mel-spectrogram, phase, and onset-weighted losses with physics regularisers; trained on MAESTRO via Modal cloud GPUs.
- Companion 8-voice polyphonic JUCE/C++ VST integrating trained parameters in real time via structured JSON.

Technical Skills

Languages: Python, TypeScript, JavaScript, Go, Lua, Bash, Java, C, C++

AI / ML: PyTorch, TensorFlow, GANs, Diffusion Models, Differentiable DSP, ONNX Runtime, Ray Serve, MLflow, Feast, Evidently, Pipecat

LLM Engineering: OpenAI / Claude / Gemini SDKs, tool use & agents, prompt registries, AI Gateway, evals

Audio / DSP: MIDI, Synthesis, Physical Modelling, JUCE / VST development, Reaper / ReaScript

Web / Backend: Hono, FastAPI, Django, Flask, HTMX, Svelte, Tailwind / daisyUI, Bun, Drizzle

Infra: Cloudflare Workers / D1 / R2 / Durable Objects, Apache Flink, Redpanda / Kafka, Redis, Docker, Kubernetes (Helm), Terraform, Prometheus / Grafana

Tools: Git, Linux, macOS

Education

University of Glasgow — BSc (Hons) Computer Science

Expected June 2026

Relevant coursework: Artificial Intelligence, Machine Learning, Deep Learning (MSc-level), Text as Data, Big Data.